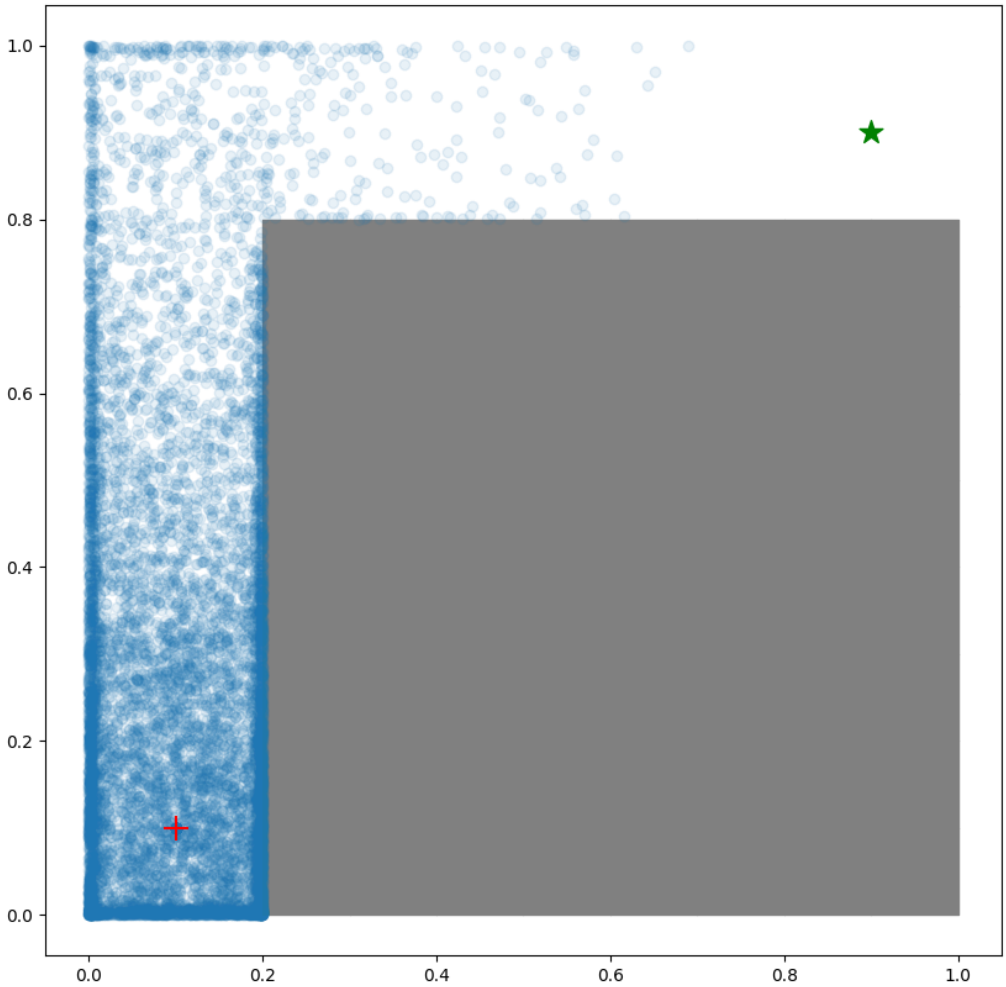
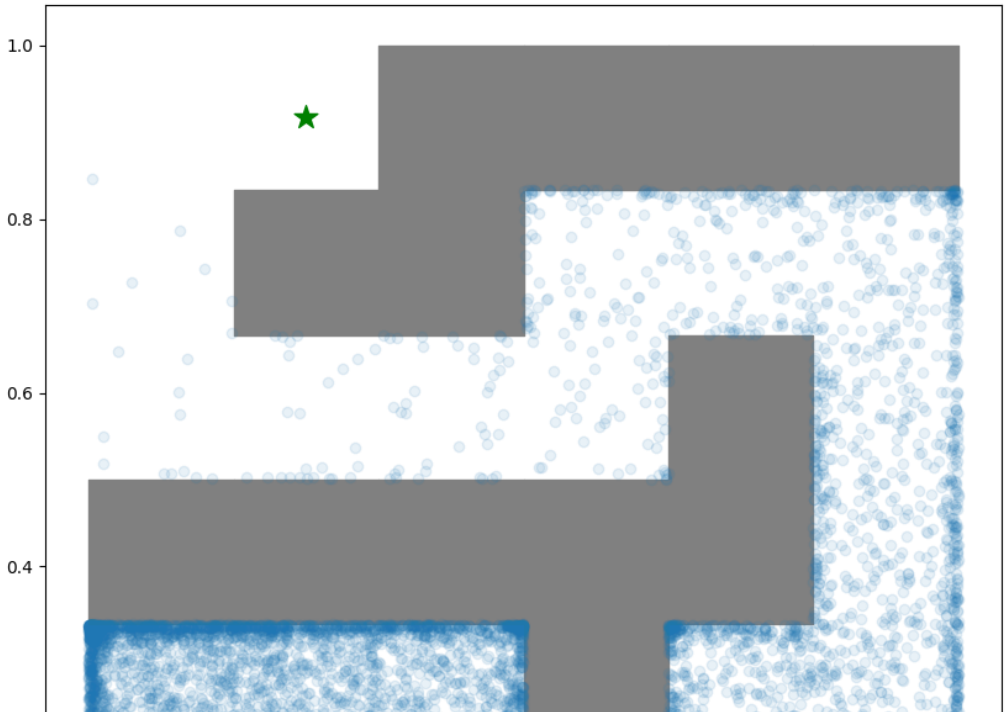
Exploration

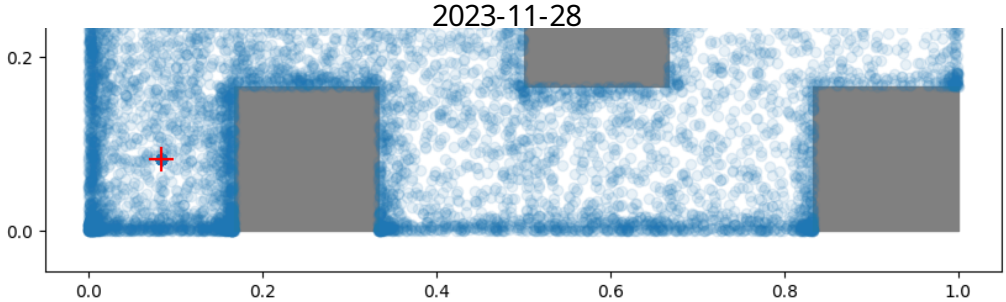
random Agent

State coverage

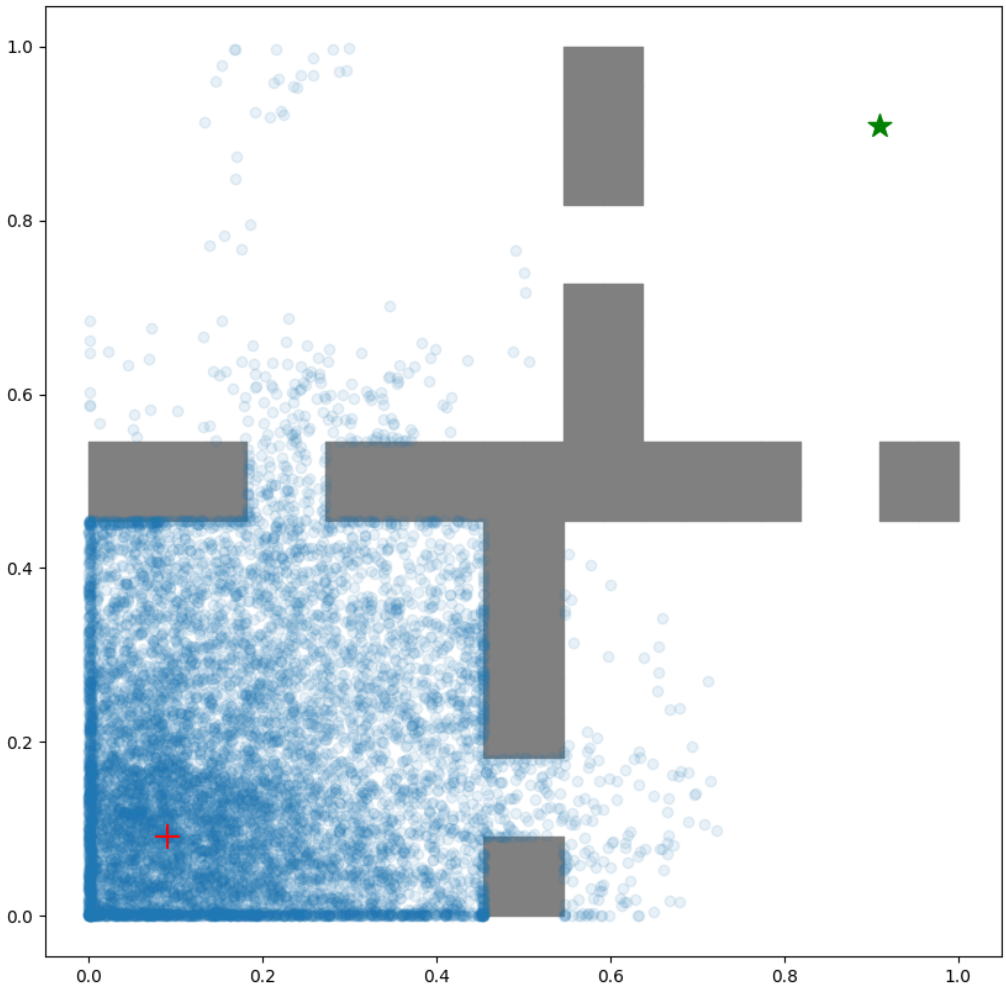


State coverage





State coverage



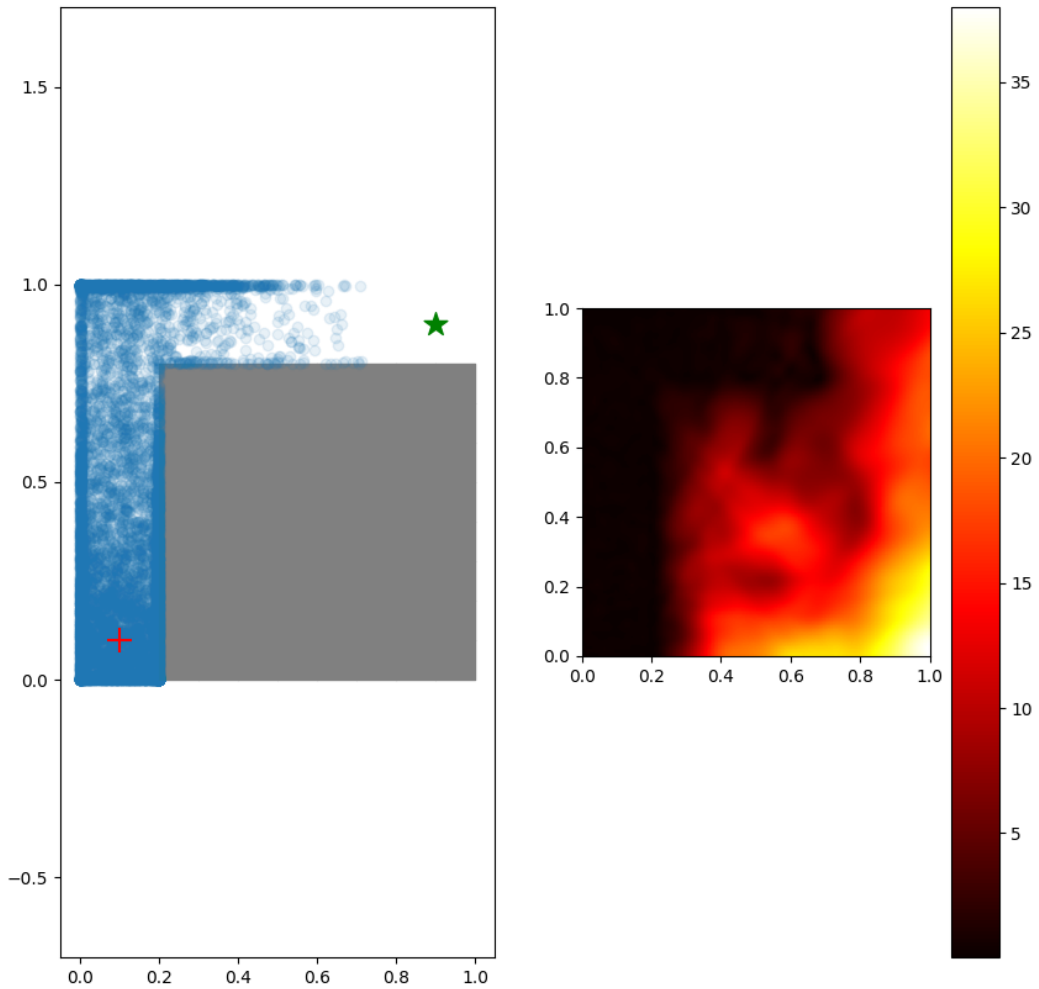
rnd Agent

The Random Network Distillation algorithm encourages exploration by training another neural network to approximate the output of a randomly initialized target network, using the discrepancy in predictions to explore new state-action spaces.

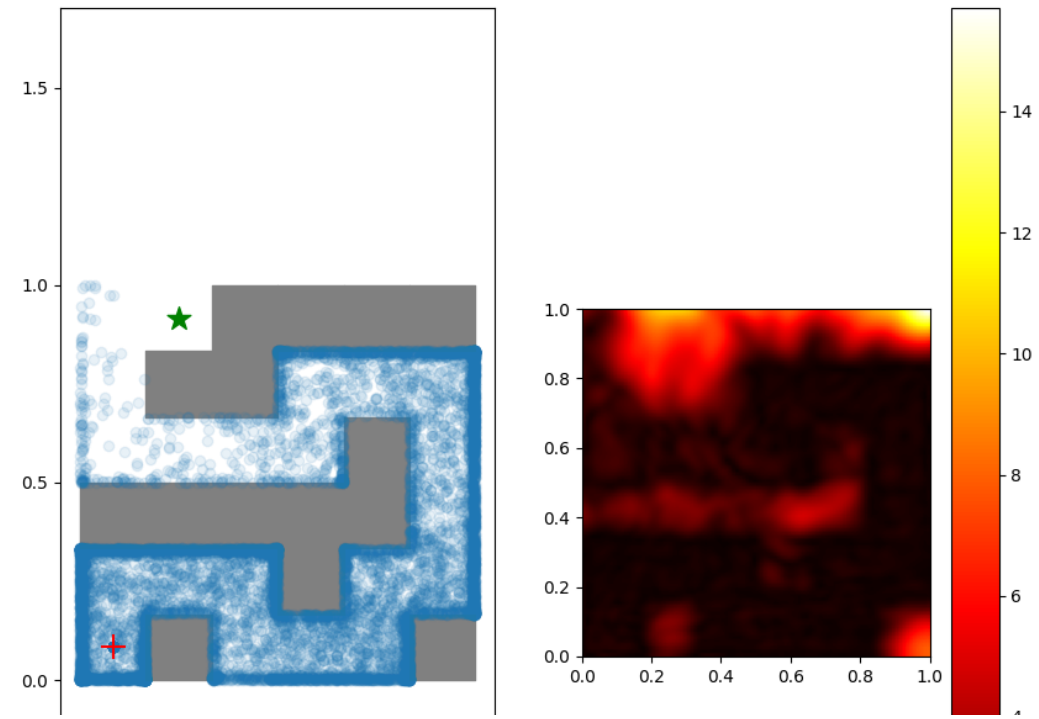
$$\phi^* = \arg\min_{\phi} \mathrm{E}_{s,a,s' \sim D} \left[\left\| \hat{f}_{\phi}(s') - f_{\theta^*}(s') \right\| \right]$$

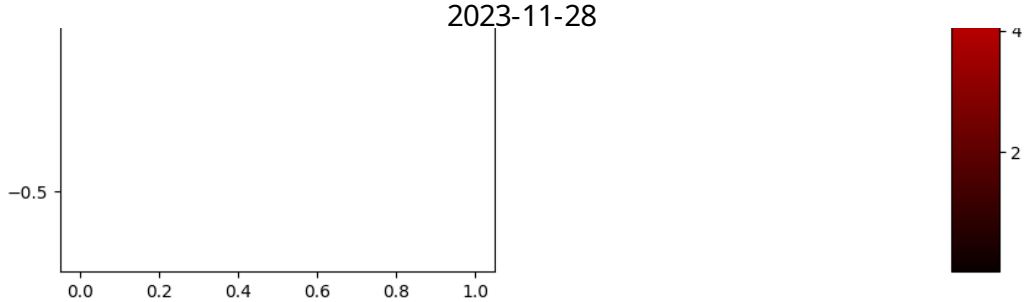
▶ Details

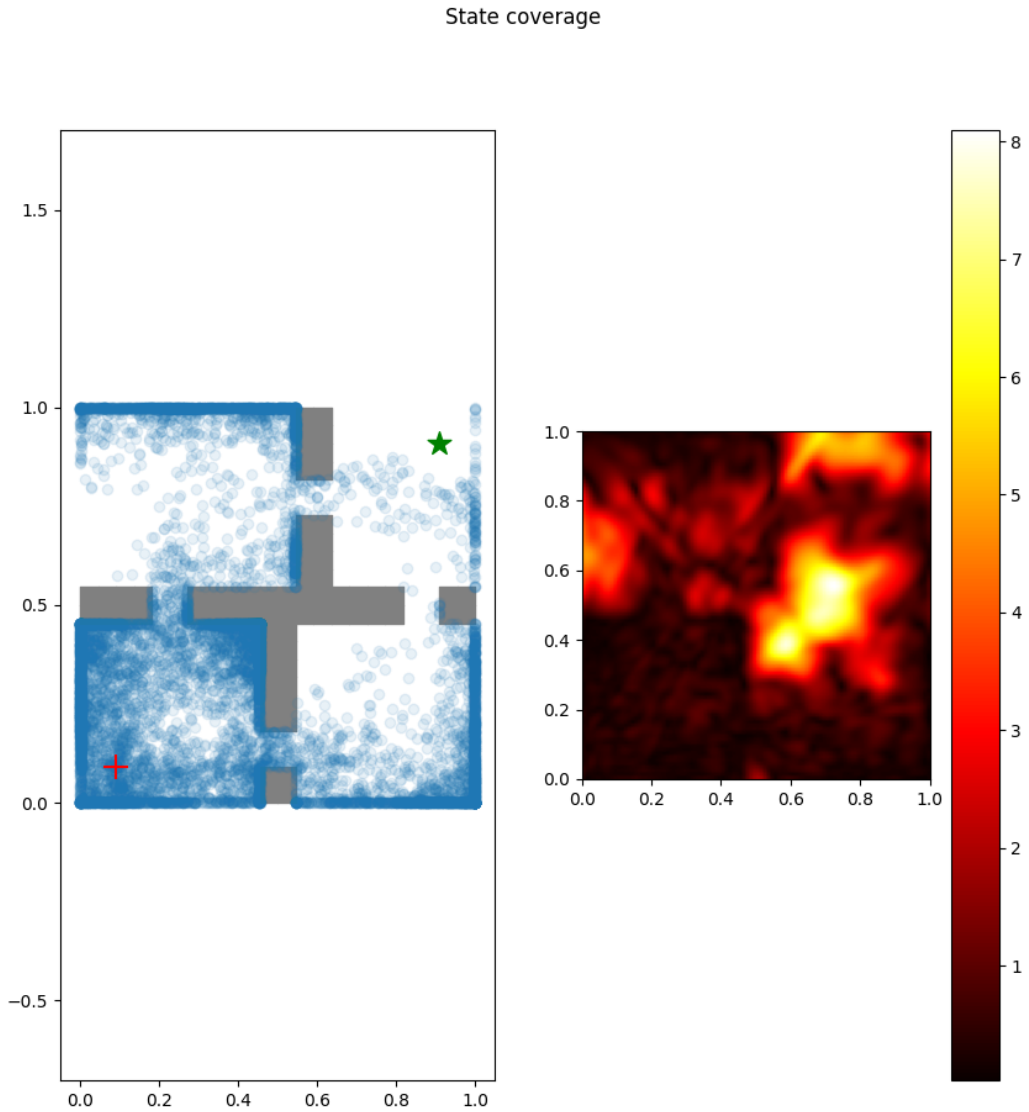
State coverage



State coverage

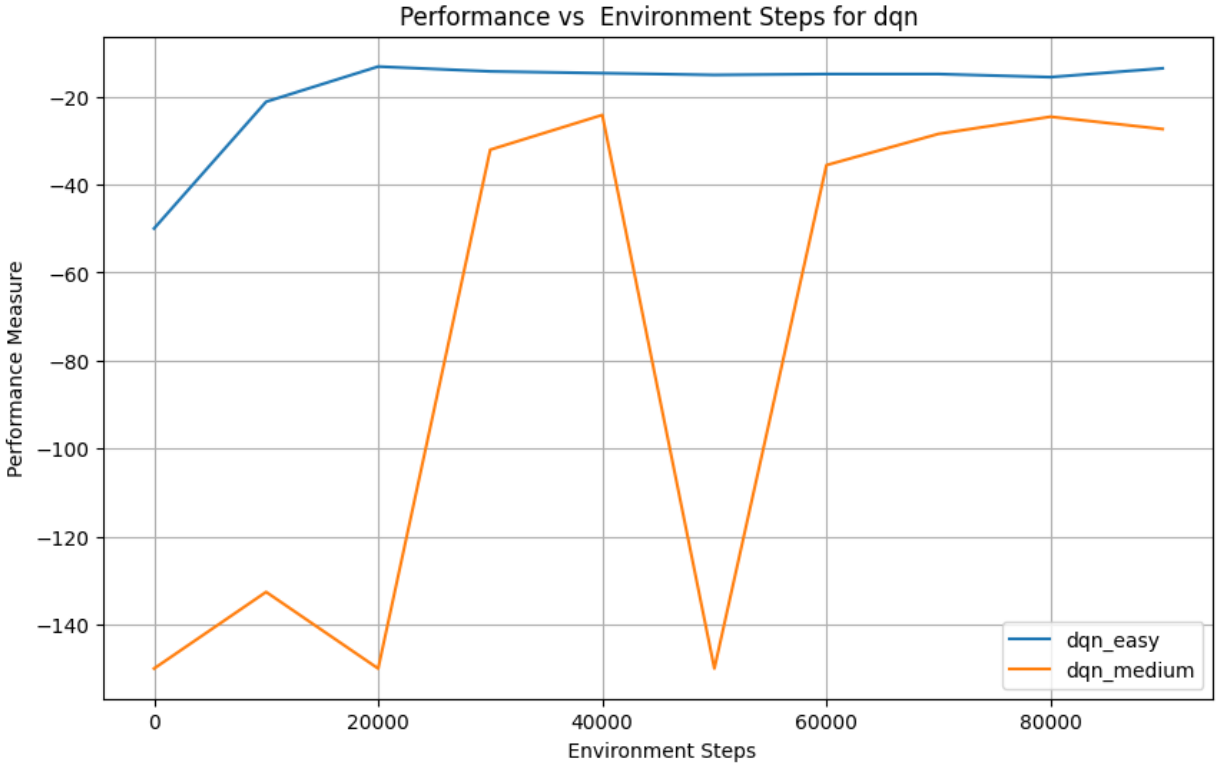


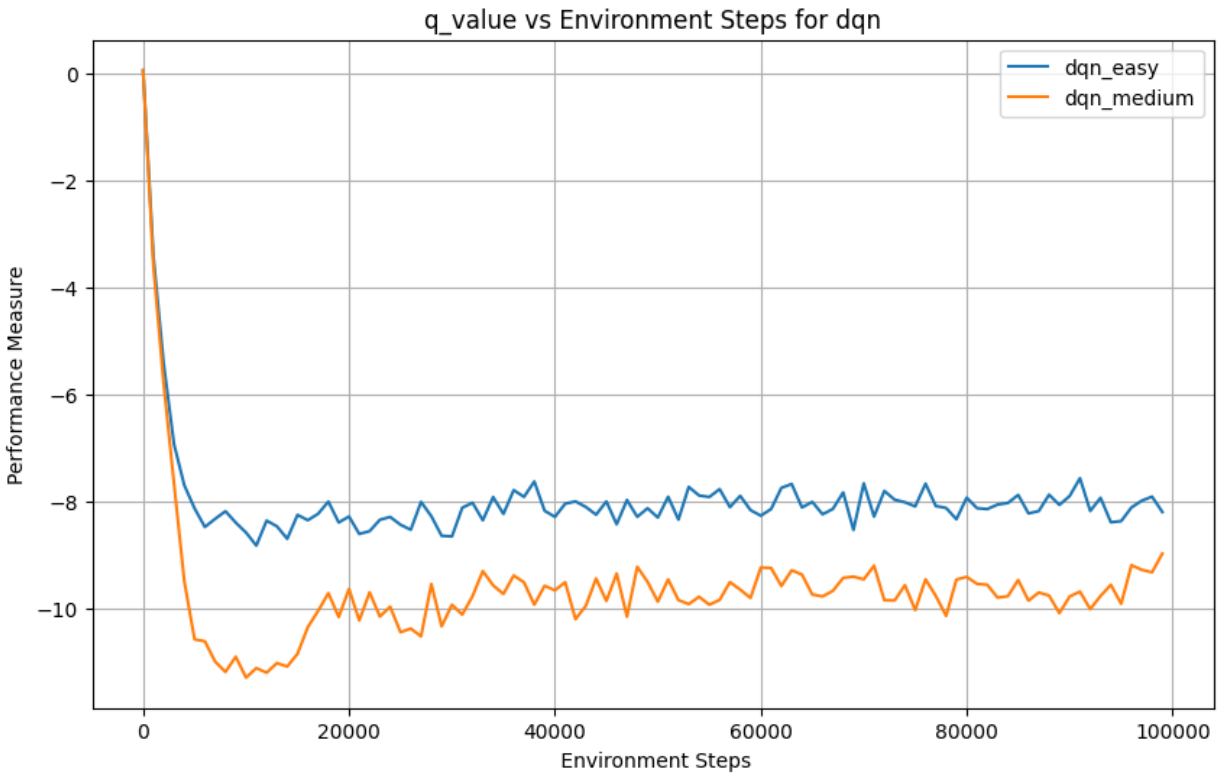




Offline RL

dqn Agent



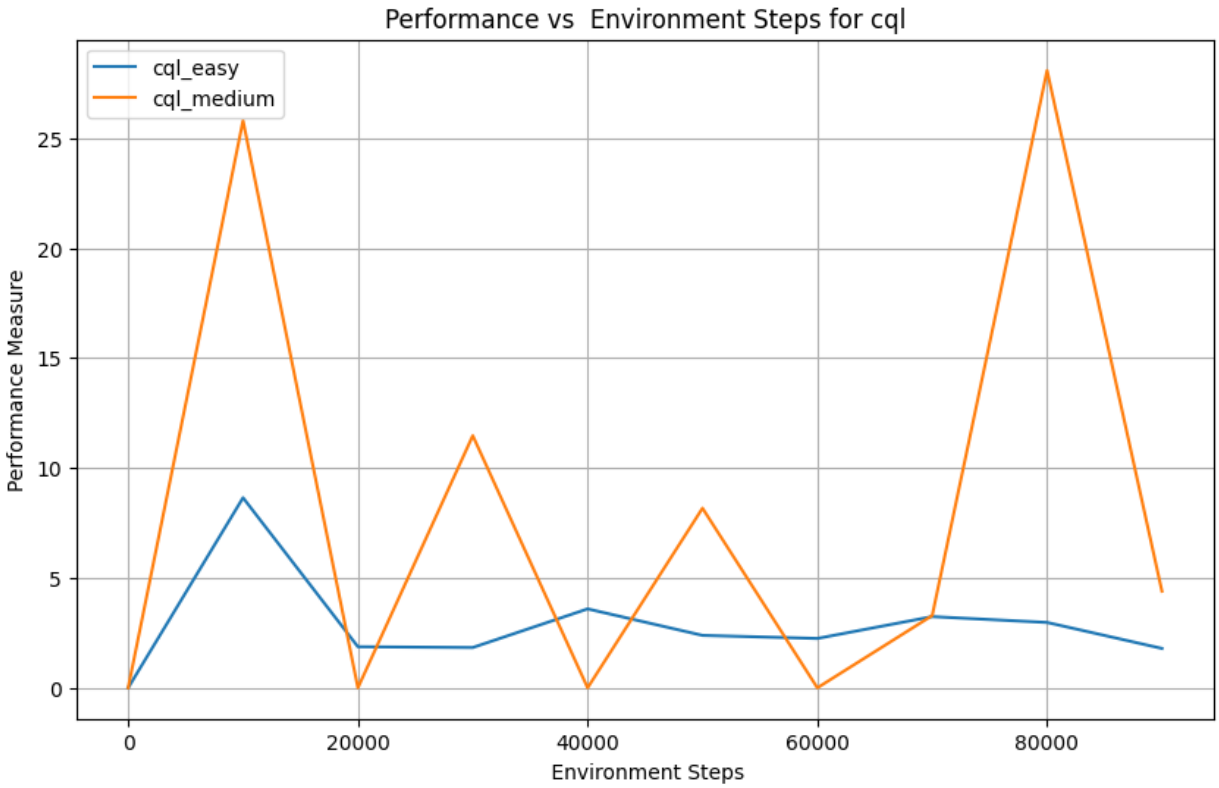


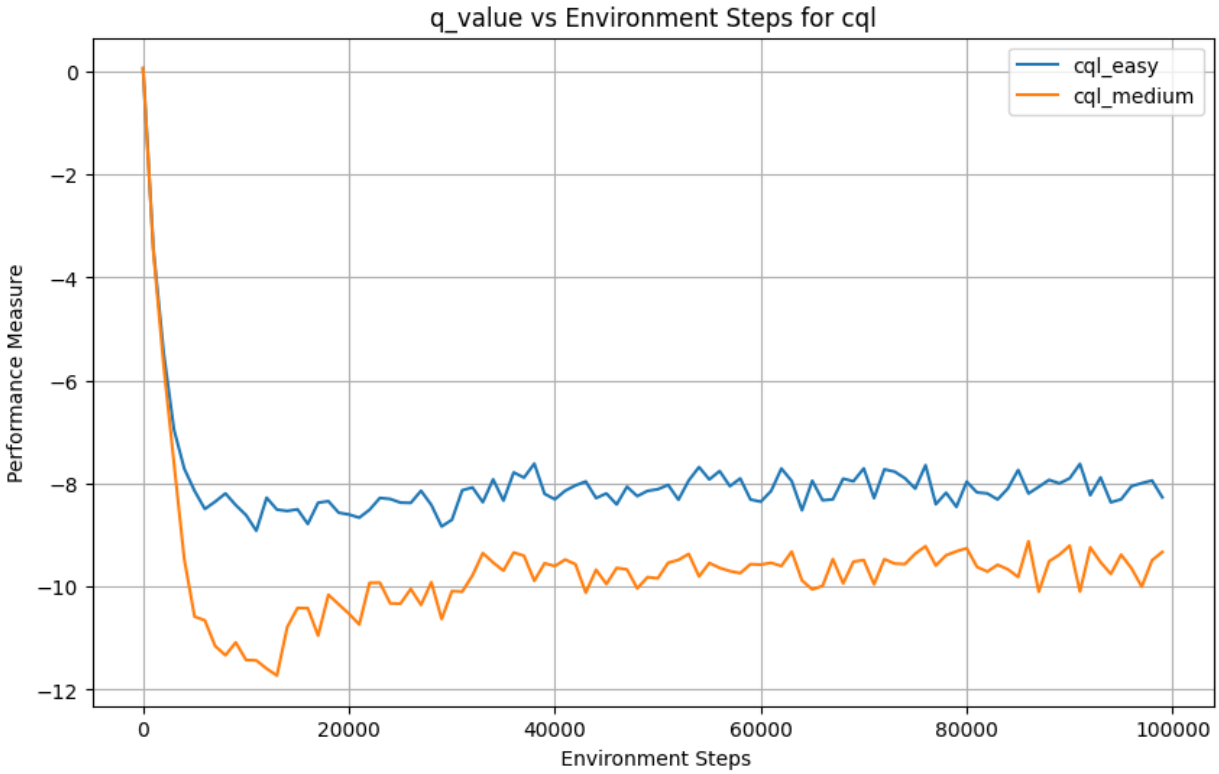
cql Agent

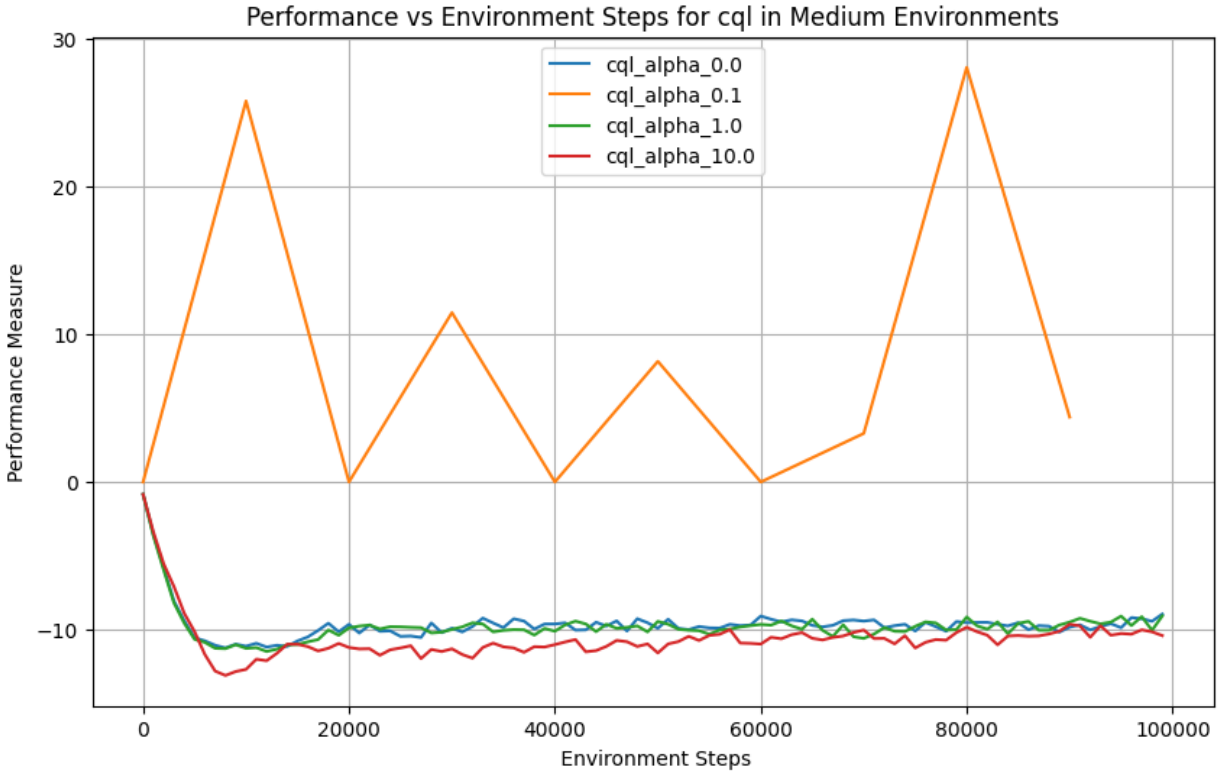
Conservative Q-Learning in offline reinforcement learning aims to prevent policy value overestimation by learning a lower-bound Q-function, reducing those for unseen state-action pairs.

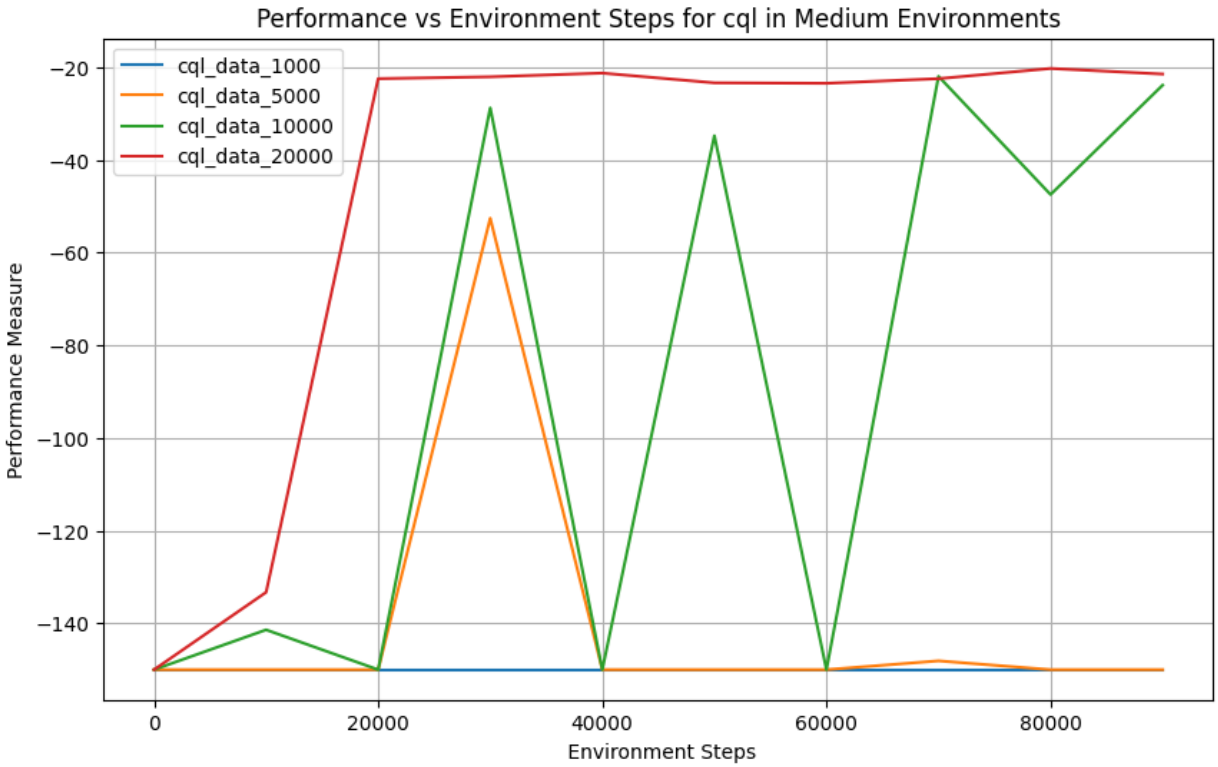
$$\text{TD Error} + \alpha \times \frac{1}{N} \sum_{i=1}^{N} (\log \sum_{a} \exp(\frac{Q(s_i, a)}{\tau}) - Q(s_i, a_i))$$

▶ Details


Performance vs Environment Steps for cql
cql_easy
cql_medium
Performance Measure
Environment Steps
0
5
10
15
20
25
0
20000
40000
60000
80000


q_value vs Environment Steps for cql
cql_easy
cql_medium
Performance Measure
Environment Steps
0
−2
−4
−6
−8
−10
−12
0
20000
40000
60000
80000
100000





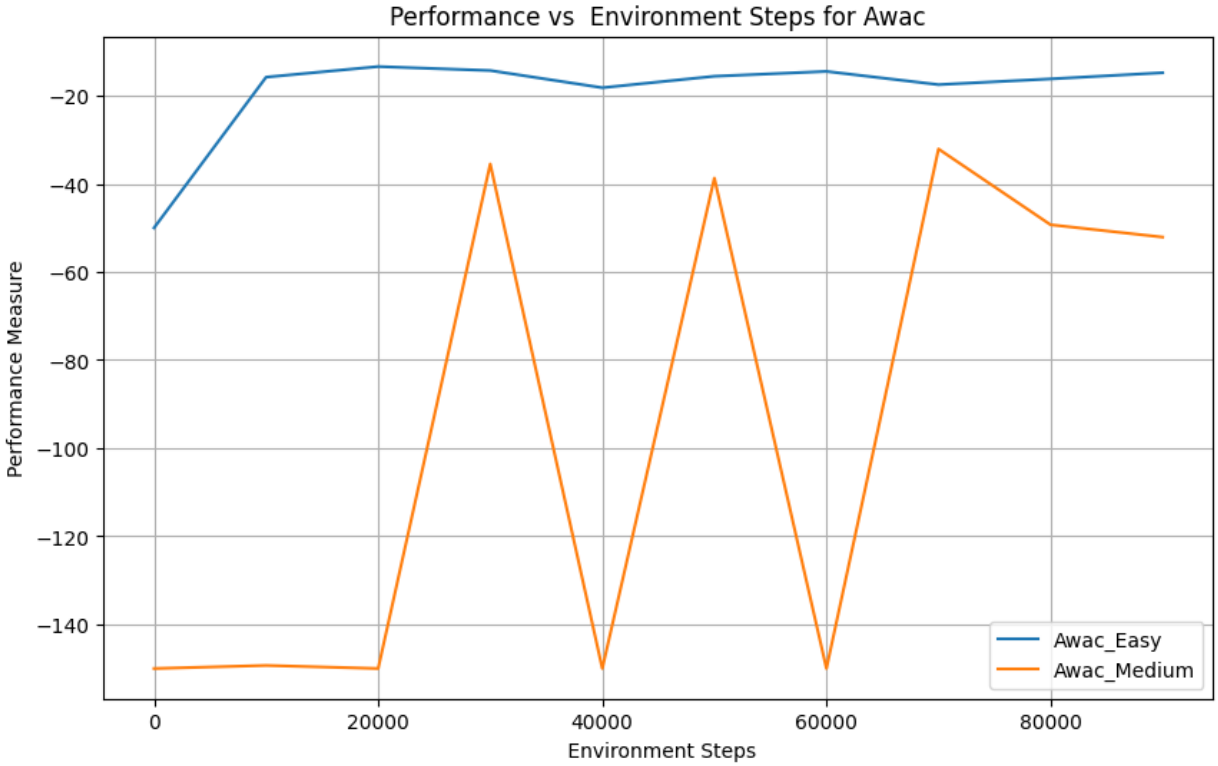
awac Agent

$$\mathrm{E}_{\mathrm{D}}\left[(Q(s,a)-r(s,a)+\gamma \mathrm{E}_{a^{'}\sim\pi}\left[Q_{\phi_{k-1}}(s^{'},a^{'})\right])^{2}\right] \tag{7}$$

▶ Details

$$\theta \leftarrow \arg\max_{\theta} \mathrm{E}_{s,a\sim \mathrm{B}}\left[\log \pi_{\theta}(a \mid s) \exp\left(\frac{1}{\lambda} A^{\pi_k}(s,a)\right)\right]. \tag{6}$$

▶ Details



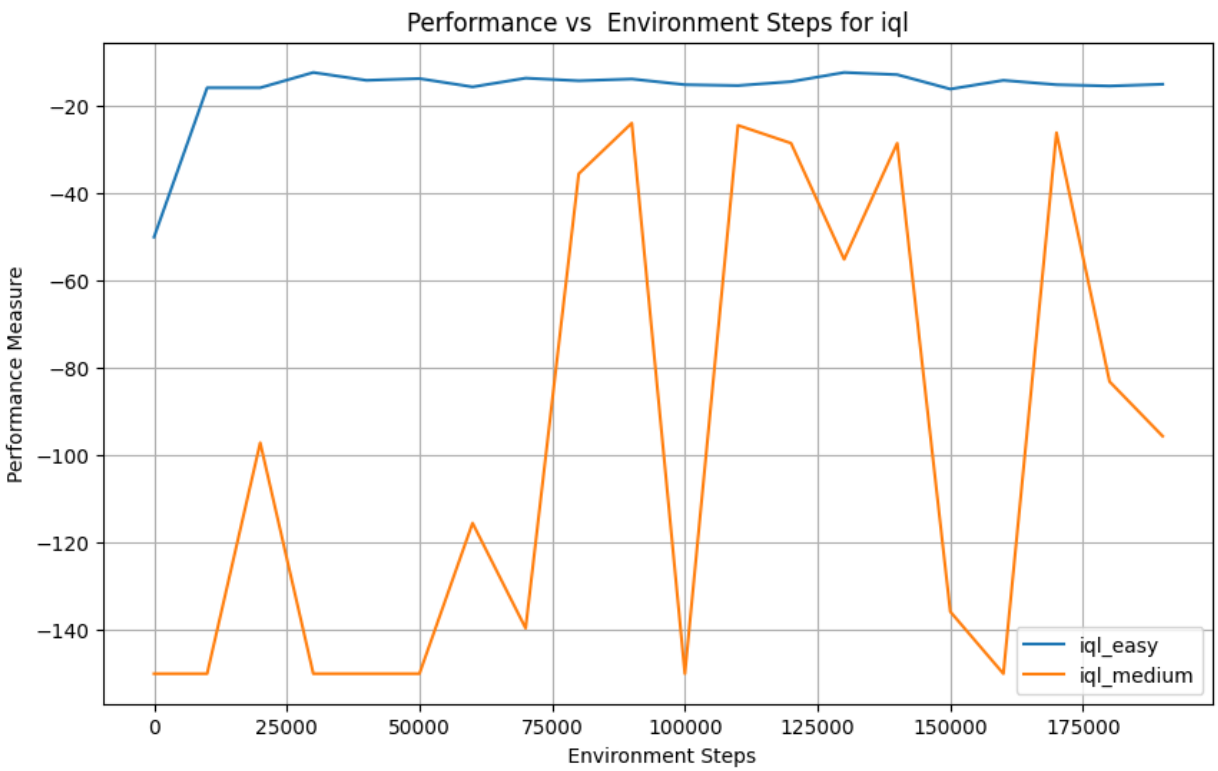
iql Agent

$$L_Q(\theta) = \mathrm{E}_{(s,a,s')\sim D}[(r(s,a) + \gamma V_\phi(s') - Q_\theta(s,a))^2]$$

▶ Details

$$L_V(\phi) = \mathrm{E}_{(s,a)\sim D}[L_\tau^2(Q_\theta(s,a) - V_\phi(s))]$$

▶ Details



Experiements to run

```
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_easy_random.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_medium_random.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_hard_random.yaml \
--dataset_dir datasets/ --log_interval 1000


python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_easy_rnd.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_medium_rnd.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_hard_rnd.yaml \
--dataset_dir datasets/ --log_interval 1000



#4.1

python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_easy_cql.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \

python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_alpha0.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_alpha10.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_alpha1.yaml \
--dataset_dir datasets --log_interval 1000


python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_easy_dqn.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_dqn.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
```

```
#4.2

python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_easy_awac.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_awac.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \


python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_easy_iql.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_iql.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \




#4.3


python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_medium_rnd_1000.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_medium_rnd_5000.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_medium_rnd_10000.yaml \
--dataset_dir datasets/ --log_interval 1000
python cs285/scripts/run_hw5_explore.py \
-cfg experiments/exploration/pointmass_medium_rnd_20000.yaml \
--dataset_dir datasets/ --log_interval 1000

python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_rnd1000.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_rnd5000.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_rnd10000.yaml \
--dataset_dir datasets --log_interval 1000
python ./cs285/scripts/run_hw5_offline.py \
-cfg experiments/offline/pointmass_medium_cql_rnd20000.yaml \
--dataset_dir datasets --log_interval 1000
```